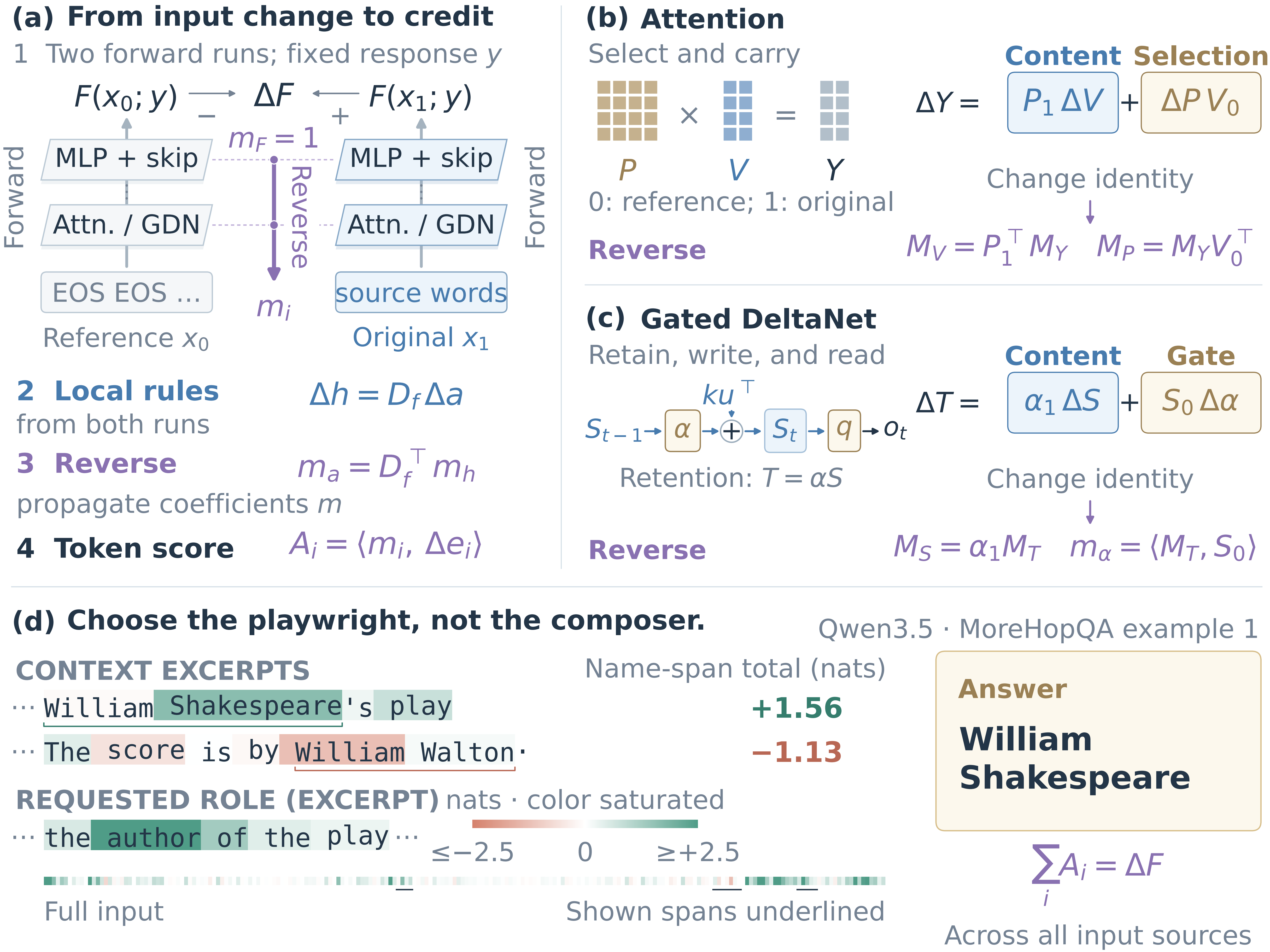

(a) From input change to credit
1 Two forward runs; fixed response y
$F(x_0; y) \rightarrow \Delta F \leftarrow F(x_1; y)$
− +
Forward
MLP + skip
Attn. / GDN
EOS EOS ...
Reference x_0
$m_F = 1$
Reverse
m_i
MLP + skip
Attn. / GDN
source words
Original x_1
Forward
2 Local rules
from both runs
$\Delta h = D_f \Delta a$
3 Reverse
propagate coefficients m
$m_a = D_f^\top m_h$
4 Token score
$A_i = \langle m_i, \Delta e_i \rangle$
(b) Attention
Select and carry
$P \times V = Y$
0: reference; 1: original
Content Selection
$\Delta Y = P_1 \Delta V + \Delta P V_0$
Change identity
Reverse
$M_V = P_1^\top M_Y$ $M_P = M_Y V_0^\top$
(c) Gated DeltaNet
Retain, write, and read
$ku^\top$
$S_{t-1} \rightarrow \alpha \rightarrow \oplus \rightarrow S_t \rightarrow q \rightarrow o_t$
Retention: $T = \alpha S$
Content Gate
$\Delta T = \alpha_1 \Delta S + S_0 \Delta \alpha$
Change identity
Reverse
$M_S = \alpha_1 M_T$ $m_\alpha = \langle M_T, S_0 \rangle$
(d) Choose the playwright, not the composer.
Qwen3.5 · MoreHopQA example 1
CONTEXT EXCERPTS
Name-span total (nats)
··· William Shakespeare's play
+1.56
··· The score is by William Walton·
−1.13
REQUESTED ROLE (EXCERPT) nats · color saturated
··· the author of the play ···
≤−2.5 0 ≥+2.5
Full input
Shown spans underlined
Answer
William Shakespeare
$\sum_i A_i = \Delta F$
Across all input sources